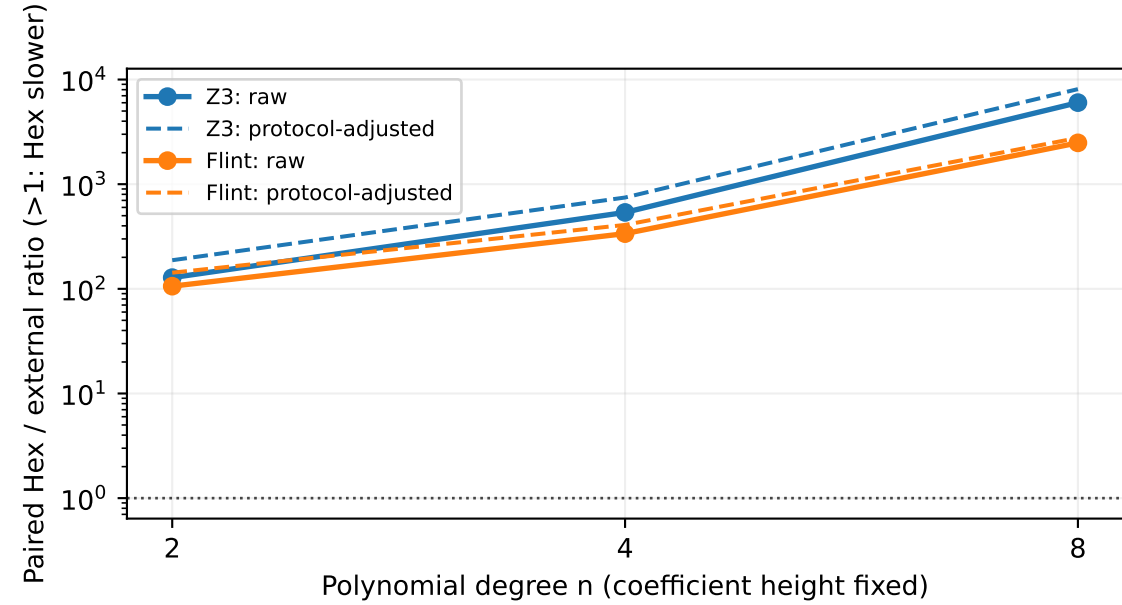
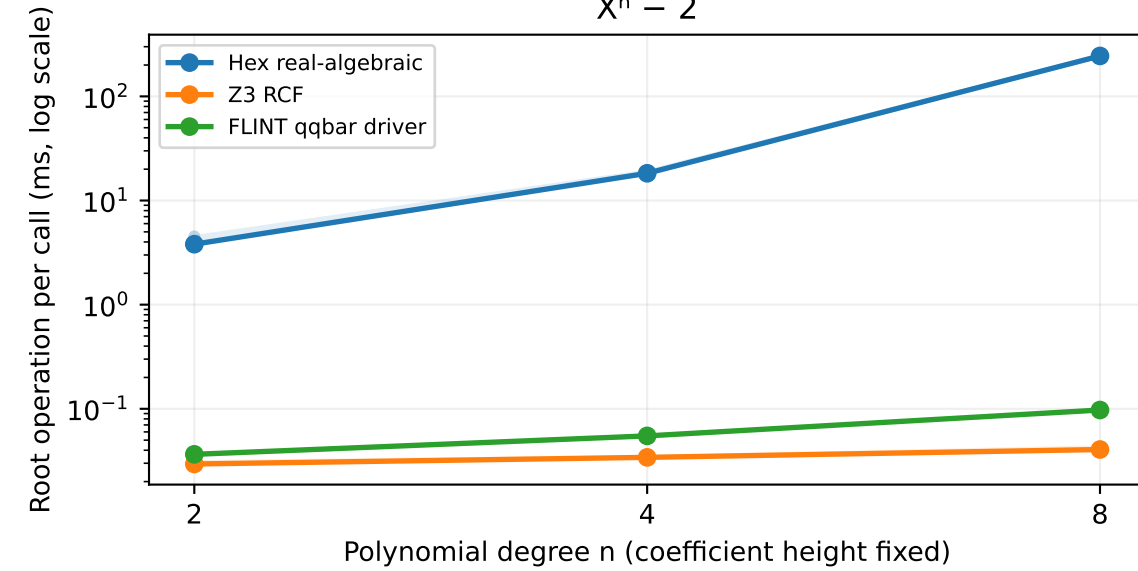
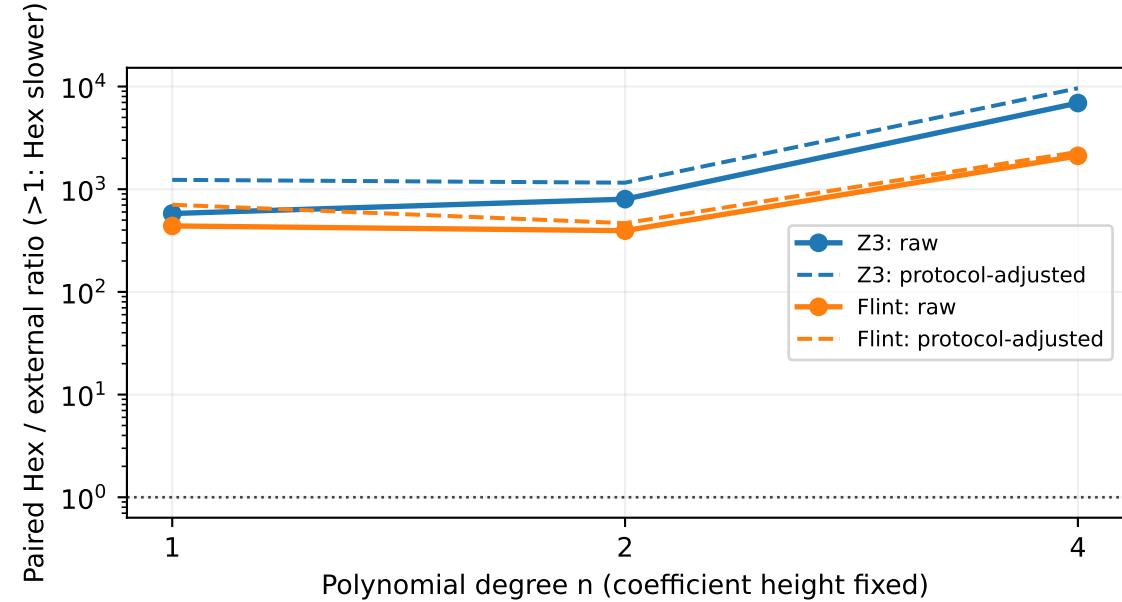
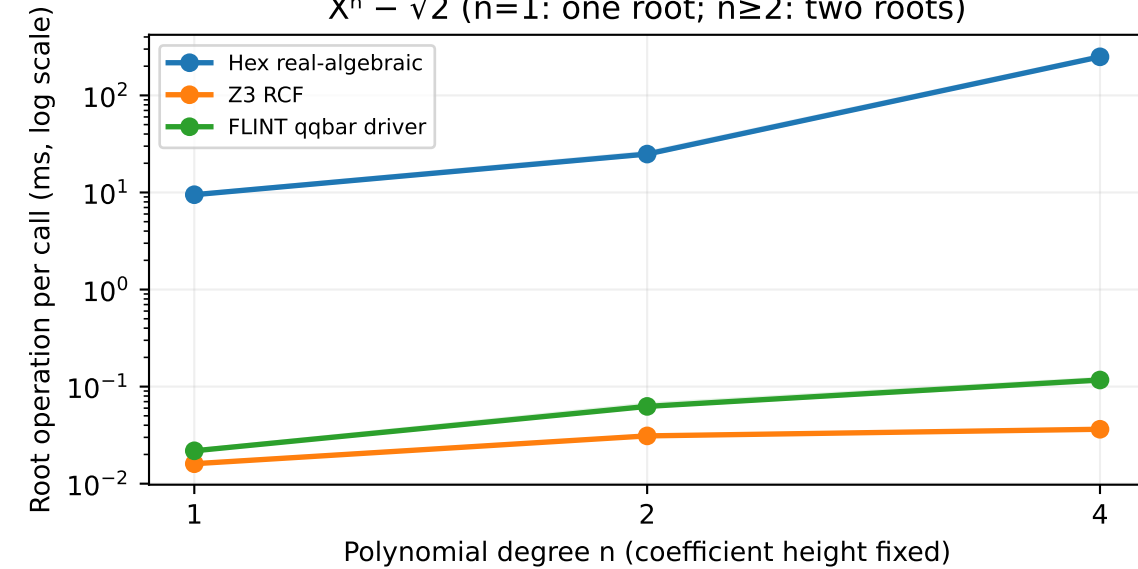


After early nonreal rejection: Hex roots, Z3 RCF and FLINT qqbar

$$X^n - 2$$



$$X^n - \sqrt{2} \text{ (n=1: one root; n} \geq 2 \text{: two roots)}$$



Shared host chungus2, leased CPU 85; 4 adjacent AB/BA pairs per backend/rung. All samples retained; no fitted model. Prepared coefficients; solving, ordering and fingerprints timed. External exact-annihilation checks, JSON and cleanup included. Separate unpinned degree-16 rational and degree-8 quadratic probes hit a 60 s whole-child cap; no numerical operation time inferred. Source edb3ef956; protocol adjustment is a framing control. Comparison failures/censored arms: 0; collected 144/144.